

ROS2-Simox Grasp Planning Roadmap

Current Status

- ROS2 package exists.
- Basic functionality for receiving object and hand data via ROS.
- Single grasp returned from Simox API.

Phase 1 – Core Integration & API Development

Goal: Build a robust ROS2-Simox bridge with clean data handling.

1. ROS2 Service/Action Interface

- Implement `/plan_grasps` service:
 - Inputs: object mesh & pose, hand model, number of grasps to return.
 - Outputs: list of grasp poses, grasp quality scores, success flag.

2. Data Conversion Utilities

- Create `ros_to_simox` and `simox_to_ros` helpers for:
 - Pose, mesh, and transform conversions.
 - Grasp quality metric extraction.

3. Error Handling

- Add descriptive error messages for:
 - Missing/invalid mesh data.
 - Simox API call failures.
 - No feasible grasp found.

Phase 2 – Grasp Quality & Reachability

Goal: Return only grasps that the robot can actually execute.

1. Reachability Filtering

- Use Virtual Robot IK solvers to discard unreachable grasps.

2. Collision Avoidance

- Integrate Saba collision checking for both environment and self-collisions.

3. Ranking Grasps

- Sort candidates by Simox grasp quality score.

Phase 3 – Visualization & Debugging Tools

Goal: Make results easy to interpret.

1. RViz2 Visualization

- Publish grasps as `visualization_msgs::MarkerArray`:
 - Arrows for approach direction.
 - Color coding based on quality score.

2. Simox Visualization

- Enable Coin3D/SoQt rendering for debugging in Simoxs native viewer.

3. Logging

- Save planned grasps, mesh files, and metrics to disk for analysis.

Phase 4 – Testing, Validation, and Benchmarking

Goal: Ensure reliability across various scenarios.

1. Unit Tests

- Test ROS–Simox conversions.
- Mock Simox calls to verify service logic.

2. Benchmark Dataset

- Use standard test objects (box, cylinder, mug, etc.).
- Measure planning time, success rate, and grasp quality.

Phase 5 – Advanced Capabilities

Goal: Move toward a deployable real-world grasp planning system.

1. Dynamic Environment Updates

- Update Simox scene when new obstacles are detected via sensors.

2. Batch Grasp Planning

- Plan grasps for multiple objects in one request.

3. Multi-Hand / Multi-Robot Support

- Allow selecting different robot arms or grippers dynamically.

4. **Real-World Validation**

- Deploy on a physical robot, compare simulation vs. execution success.